

Understanding DRC & LVS of SPM Design Using Openlane

INTRODUCTION:

Design Rule Check (DRC) and Layout Versus Schematic (LVS) are critical signoff verification steps in the RTL-to-GDSII flow. After routing is completed in OpenLane, the physical layout must strictly follow the foundry design rules and must be electrically equivalent to the synthesized netlist. OpenLane integrates open-source tools such as Magic and Netgen to perform DRC and LVS verification. DRC ensures that the layout complies with manufacturing constraints, while LVS confirms that the layout connectivity matches the logical design. Successful DRC and LVS checks indicate that the design is fabrication-ready.

APPLICATION:

The DRC and LVS verification stage in OpenLane is used to validate the physical correctness of digital ASIC designs such as processors, controllers, and SoCs. It helps designers identify design rule violations, layout connectivity errors, missing or extra devices, and mismatches between the schematic and the layout. DRC and LVS are widely applied in educational ASIC flows, RISC-V cores, and open-source silicon implementations using Sky130 and GF180 PDKs. Proper DRC and LVS verification ensures manufacturability, electrical correctness, and readiness of the design for final physical signoff and GDSII.

OBJECTIVE:

After completing this lab, you will be able to:

- Understand the importance of DRC and LVS in the ASIC design flow
- Perform DRC checks using Magic in OpenLane
- Perform LVS verification using Netgen in OpenLane
- Analyse and verify DRC and LVS reports
- Prepare the design for signoff and GDSII generation

PROCEDURE TO CREATE A LAB:

Step 1: Set up OpenLane flow

1.1: Invoke OpenLane in the terminal using the command: `openlane`

```
coe-4@ip-10-100-19-29:~/OpenLaneUser/designs$ openlane
=====
OpenLane started for user: coe-4
Workspace : /home/coe-4/OpenLaneUser
PDK Root  : /labroot/openlane_pdk
=====
OpenLane Container (ff5589f):/openlane$
```

1.2: Open folder- designs: `cd designs`

Step 2: Set up Project Directory

2.1: Create a directory for your lab: `mkdir <project_name>`

2.2: Open lab directory: `cd <project_name>`

2.3: Create directory for project: `mkdir spm`

2.4: Open project directory: `cd spm`

2.5: Inside project directory, create folder: `mkdir src`

2.6: Open folder: `cd src`

Step 3: Create Design file

3.1: Open vi editor for design with command: `vi spm.v`

Note: for RTL code refer the previous document of Demonstration of floorplan using openlane

Step 4: Create .json file

4.1: Open vi editor: `vi config.json`

```
{
    "DESIGN_NAME": "spm",
    "VERILOG_FILES": "dir::src/*.v",
    "CLOCK_PERIOD": 10,
    "CLOCK_PORT": "clk",
    "CLOCK_NET": "ref::$CLOCK_PORT",
    "FP_PDN_VOFFSET": 7,
    "FP_PDN_HOFFSET": 7,
    "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
    "FP_PDN_SKIPTRIM": true,
    "pdk::sky130*": {
        "FP_CORE_UTIL": 45,
```

```
"scl::sky130_fd_sc_hd": {  
    "CLOCK_PERIOD": 10  
},  
"scl::sky130_fd_sc_hd11": {  
    "CLOCK_PERIOD": 10  
},  
"scl::sky130_fd_sc_hs": {  
    "CLOCK_PERIOD": 8  
},  
"scl::sky130_fd_sc_ls": {  
    "CLOCK_PERIOD": 10,  
    "MAX_FANOUT_CONSTRAINT": 5  
},  
"scl::sky130_fd_sc_ms": {  
    "CLOCK_PERIOD": 10  
}  
},  
"pdk::gf180mcu*": {  
    "CLOCK_PERIOD": 24.0,  
    "FP_CORE_UTIL": 40,  
    "MAX_FANOUT_CONSTRAINT": 4,  
    "PL_TARGET_DENSITY": 0.5  
}  
}
```

Step 6: Use the following commands to run the flow:

OpenLane to start the design flow in interactive mode

i) `flow.tcl -interactive`

Used in the OpenLane interactive shell to prepare your design for the digital IC design flow.

ii) `prep -design spm`

Runs yosys synthesis on the current design.

iii) run_synthesis

To initiate floorplanning

iv) run_floorplan

To initiate placement

v) run_placement

To initiate cts

vi) run_cts

To initiate routing

vii) run_routing

To initiate magic tool

viii) run_magic

For DRC check

ix) run_magic_drc

```
% run_magic_drc
[STEP 25]
[INFO]: Running Magic DRC (log: designs/spm/runs/RUN_2026.02.02_09.18.50/logs/signoff/25-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
%
```

#For Spice extraction

x) run_magic_spice_export

For klayout

xi) run_klayout

#For LVS

xii) run_lvs

```
% run_lvs
[STEP 28]
[INFO]: Writing Powered Verilog (logs: designs/spm/runs/RUN_2026.02.02_09.18.50/logs/signoff/28-write_powered_def.log, designs/spm/runs/RUN_2026.02.02_09.18.50/logs/signoff/28-write_powered_verilog.log)...
[STEP 29]
[INFO]: Writing Verilog (log: designs/spm/runs/RUN_2026.02.02_09.18.50/logs/signoff/28-write_powered_verilog.log)...
[STEP 30]
[INFO]: Running LVS (log: designs/spm/runs/RUN_2026.02.02_09.18.50/logs/signoff/30-lvs.lef.log)...
%
```

Step 7: View The Report Files:

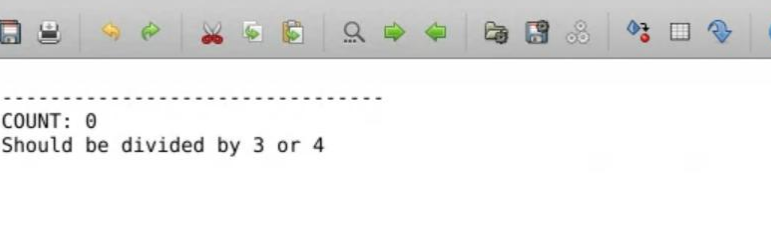
Locate the drc and the lvs report

/home/coe4/OpenLaneUser/designs/spm/runs/RUN_2026.01.30_07.13.33/reports/signoff/

```
coe-4@ip-10-100-19-29:~/OpenLaneUser/designs/spm/runs/RUN_2026.02.02_09.18.50/reports/signoff$ ls
30-spm.lvs.rpt  drc.klayout.xml  drc.rdb  drc.rpt  drc.tcl  drc.tr  spice.feedback.txt
```

Open Both the Reports with Gvim

```
gvim drc.rpt
```



drc.rpt (~ /OpenLaneUser/designs/...09.18.50/reports/signoff) - GVIM7

File Edit Tools Syntax Buffers Window Help

Spm

[INFO]: COUNT: 0

[INFO]: Should be divided by 3 or 4

<26.02.02 09.18.50/reports/signoff/drc.rpt" 5L, 99C 1,1 All

```
gvim 30-spm.lvs.rpt
```

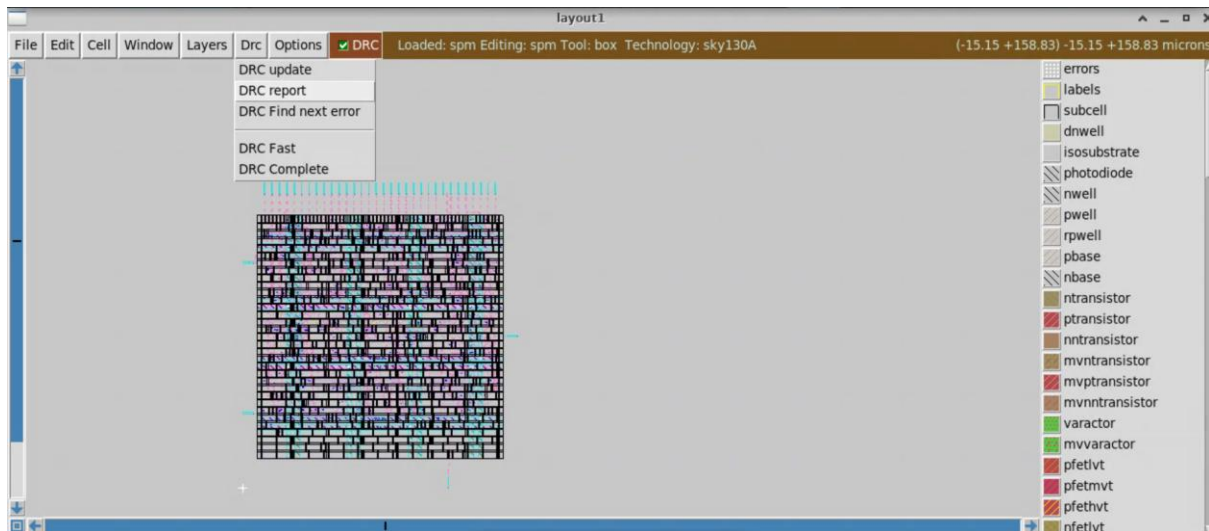
The screenshot shows a Gvim9 window titled "30-spm.lvs.rpt (~/.OpenLaneUser/d...09.18.50/reports/signoff) - GVIM9". The menu bar includes File, Edit, Tools, Syntax, Buffers, Window, and Help. The toolbar contains various icons for file operations and editing. The main text area displays the message "LVS reports no net, device, pin, or property mismatches." followed by "Total errors = 0". Below this, there are several blue tilde (~) characters representing blank lines. At the bottom status bar, it shows "<2 09.18.50/reports/signoff/30-spm.lvs.rpt" 3L, 75C" on the left, "1,1" in the center, and "All" on the right.

To see DRC in Magic tool

/home/coe4/OpenLaneUser/designs/spm/runs/RUN_2026.01.30_07.13.33/results/final/mag/spm.mag

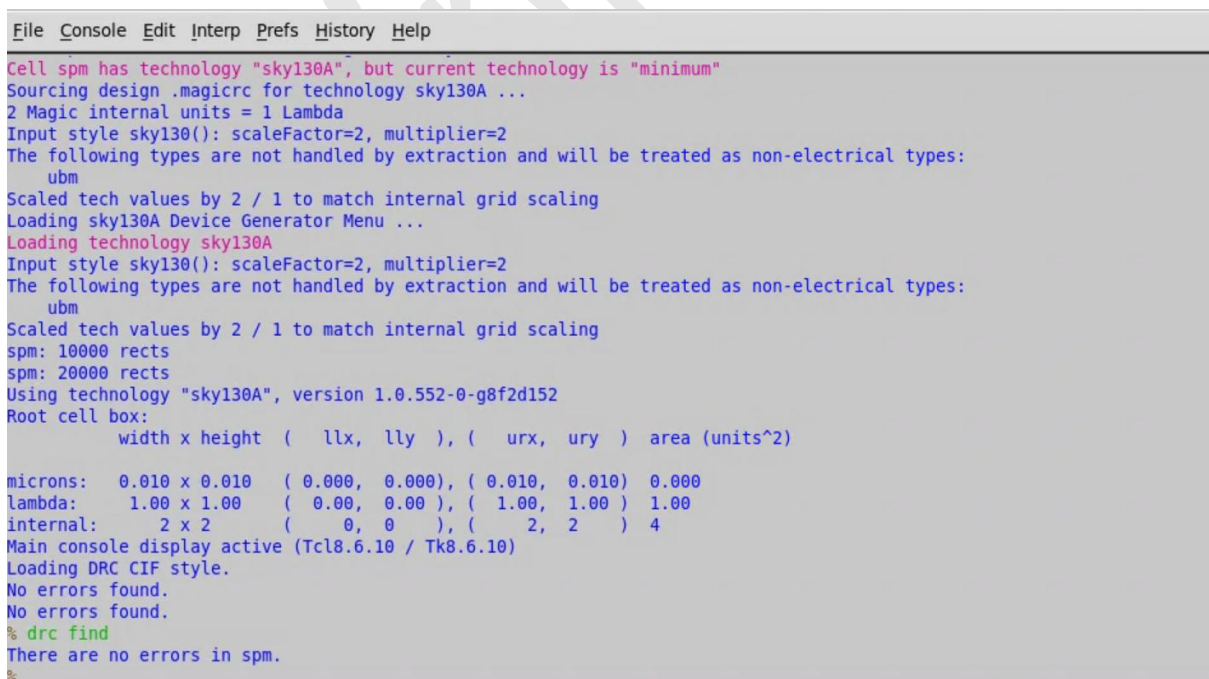
Open magic tool in the same directory where spm.def is located

i) magic576 spm.mag



Go to DRC report in the tool bar (new window comes as given below)

ii) drc find



OBSERVATION:

- DRC check completed successfully with no critical violations
- LVS comparison passed with **layout matching schematic**
- Proper metal layer usage across routing layers
- Correct via insertion between metal layers
- No open nets or short circuits detected
- Clean clock and power connectivity
- Layout verified visually using KLayout

OUTPUT:

The output of the DRC & LVS stage includes:

- DRC report (drc.rpt)
- LVS report (30-spm.lvs.rpt)
- DRC-clean routed DEF file
- Layout-extracted SPICE netlist
- Signoff-ready physical design

CONCLUSION:

In this lab, Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification were successfully performed for the SPM design using OpenLane. The physical layout adhered to all design rule constraints of the selected PDK, and the layout connectivity matched the synthesized netlist without mismatches. The successful completion of DRC and LVS confirms that the design is manufacturable, electrically correct, and ready for final signoff and GDSII generation.